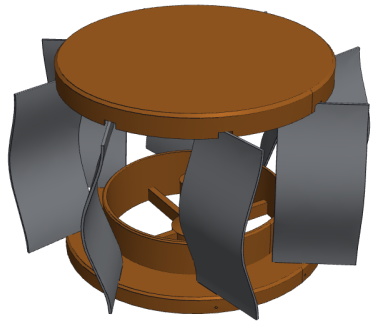
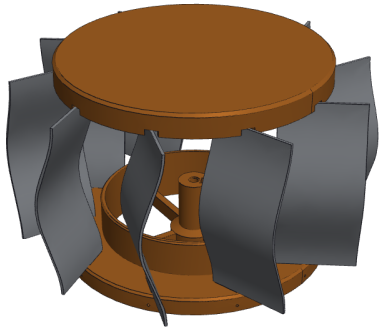
8 / 10 / 12 BLADE COMPARISON - SAME BLADE, SAME HUBS (real CAD renders)

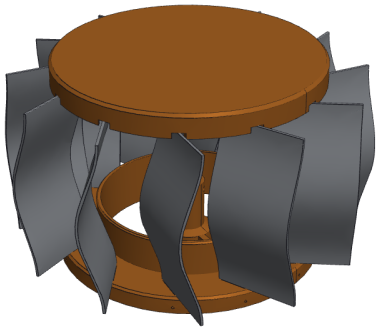
8 blades



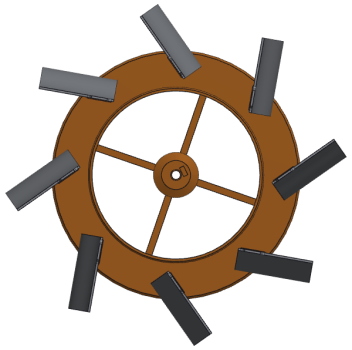
10 blades



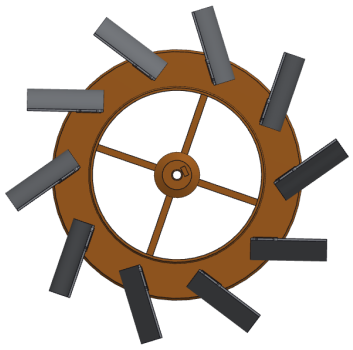
12 blades



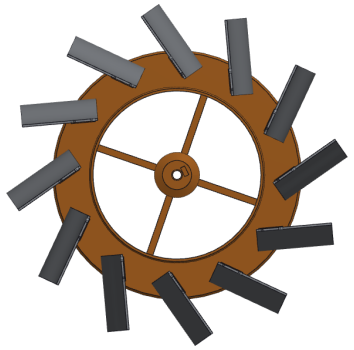
8 blades (upper hub hidden)



10 blades (upper hub hidden)



12 blades (upper hub hidden)



Blades	Chord deg	Bow	Rotor OD	Blade gap	Plan coverage %	Blade set g	Height
8	35	15	259.1	50.5	57.8	255.0	165.2
10	35	15	259.1	39.0	72.2	319.0	165.2
12	35	15	259.1	31.3	86.7	383.0	165.2

Reading the table:
- No collisions for 8, 10 or 12 with this blade envelope.
- More blades raise plan coverage (rain shading) and mass, and reduce the open exhaust area between blades.
- They do NOT guarantee more ventilation or rain-tightness.
- 8 blades remains the primary design; 10/12 are parametric (set blade_count) pending airflow tests.
- Full 27-variant table: whirlybird/reports/variant_study.csv

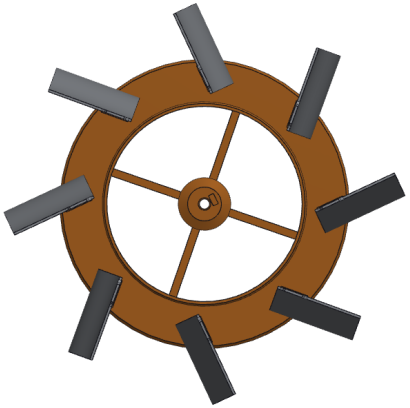
DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
8 / 10 / 12 blade comparison (real CAD)

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

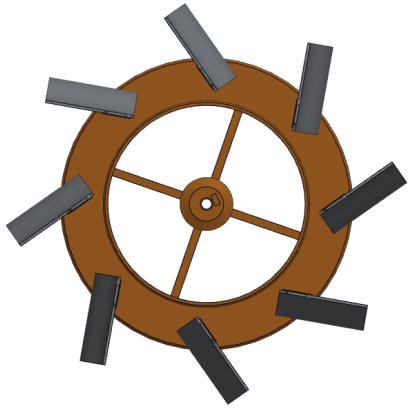
SHEET 1/2 Source: build123d parametric model + MASTER_PARAMETERS.json

PLAN CHORD ANGLE VARIANTS 25 / 35 / 45 deg (8 blades, upper hub hidden)

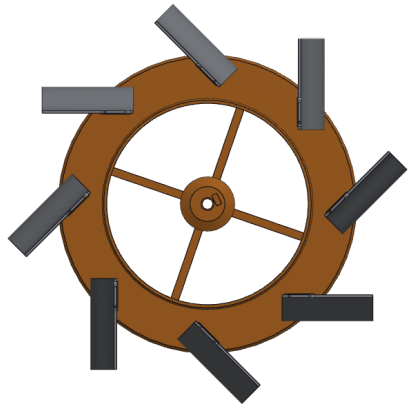
chord angle 25 deg from radial



chord angle 35 deg from radial



chord angle 45 deg from radial



Chord deg	Bow 10: OD / cov%	Bow 15: OD / cov%	Bow 20: OD / cov%	Gap mm
25	257.5 / 44.4	261.2 / 50.8	265.2 / 56.7	52.0
35	254.5 / 52.5	259.1 / 57.8	264.1 / 62.6	50.5
45	249.6 / 59.8	255.2 / 63.9	261.2 / 67.6	47.2

Selected: 35 deg, 15 mm bow - OD within the 254-279 mm window, 50 mm blade gaps, 49.6 mm minimum bend radius (easy to press-form 1.6 mm aluminium). 20 mm bow gives ~5 mm more OD but a 35 mm bend radius and more springback. 45 deg gives the most plan coverage but the smallest OD. No variant is claimed aerodynamically optimal.

DOGHOUSE WHIRLYBIRD - 8 BLADE ROTOR
Chord angle variants

REV DW-V1.1 UNITS mm SCALE NTS 2026-10-23

SHEET 2/2 Source: build123d parametric model + MASTER_PARAMETERS.json